Data Understanding: ShopSmart Retail Ltd

AnalystLab Africa Experience Lab | Week 1, Data Analytics Track | Part 1 | Divine Okezie Nwaoha

Q1. How many datasets? Ten files were provided: nine CSV datasets (Customers, Products, Orders, Order_Items, Stores, Returns, Employees, Suppliers, Inventory) and one Data Dictionary workbook. The dictionary is a reference and covers only five of the nine datasets.

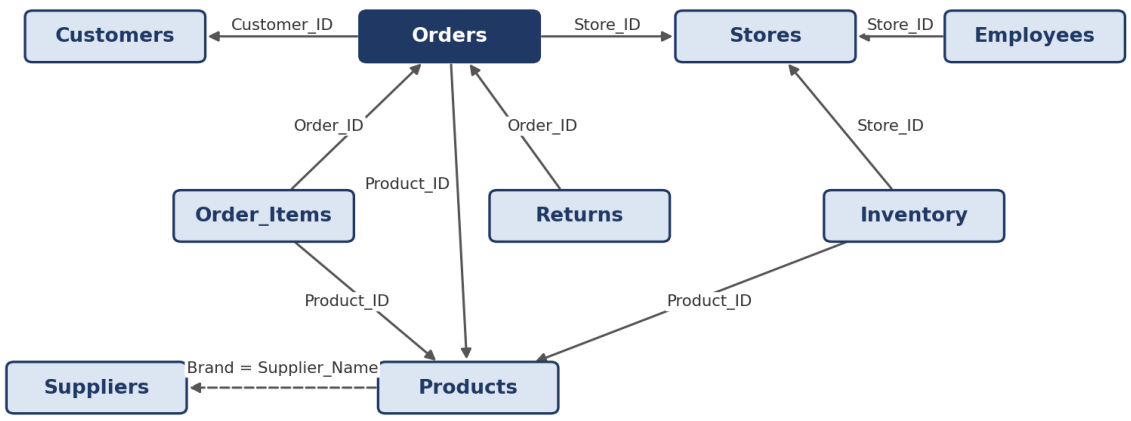
Q2 to Q4. What each dataset represents, with primary and foreign keys

Dataset	What one row represents	Primary key	Foreign key(s)
Customers	One customer: name, gender, age, city, state, registration date and customer segment (Bronze, Silver, Gold).	Customer_ID	None
Products	One product: name, category, sub-category, brand, cost price and selling price.	Product_ID	Brand matches Suppliers.Supplier_Name (see note)
Orders	One sale made by a customer, for a product, at a store, on a date, with quantity, price, discount, shipping, payment method, channel and status.	Order_ID	Customer_ID, Product_ID, Store_ID
Order_Items	One product line inside an order, with quantity, unit price, discount and line total.	Order_Item_ID	Order_ID, Product_ID
Stores	One ShopSmart store, with its region, state, city and opening date.	Store_ID	None
Returns	One product return linked to an order, with date, reason and refund amount.	Return_ID	Order_ID
Employees	One staff member, with role, store, hire date and monthly salary.	Employee_ID	Store_ID
Suppliers	One supplier company, with contact details, city and country.	Supplier_ID	None
Inventory	The stock of one product held at one store, with reorder level and last restock date.	Inventory_ID	Product_ID, Store_ID

Primary and foreign keys. A primary key is the column that uniquely identifies each row of a table (for example Customer_ID in Customers). A foreign key is a column in one table that holds the primary key of another table and so links the two (for example Customer_ID in Orders points to a customer in Customers). One customer can appear in many orders, which is why a foreign key can repeat while a primary key cannot.

Note: Products has no Supplier_ID and Suppliers has no Product_ID. Every supplier name is the product brand followed by the word “Supplies”, so products were linked to suppliers by matching Brand + “ Supplies” to Supplier_Name. Orders and Order_Items describe the same sales (same order count, matching values).

Q5. Relationship between the tables



Arrows point from the table holding the foreign key to the table it references. Orders is the central table. The dashed line is the link by name between Products and Suppliers.